UAlg FCT
UNIVERSIDADE DO ALGARVE
FACULDADE DE CIÊNCIAS E TECNOLOGIA

Generalization/Specialization (Inheritance)

Readings: ( :chapt. 12)

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Inheritance

- Inheritance is a defining feature of OOP
- Immediate issues:
  - What is Inheritance?
  - Why use it?
  - How to use it?
  - What are the other defining features of OOP, if any?

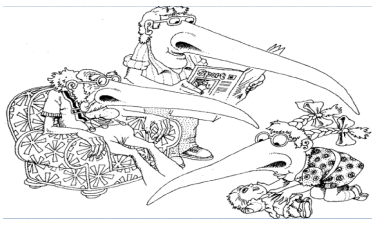


Image from: Grady Booch et al., *Object-oriented Analysis and Design with Application*, Wiley, 2007

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Example: University people



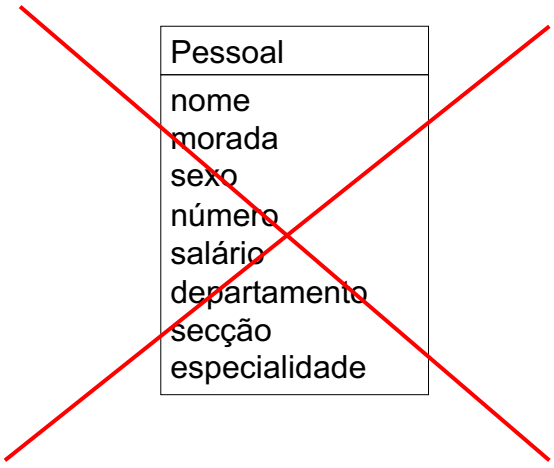
Pretende-se desenvolver um programa que guarde informação sobre as pessoas que trabalham e estudam na Universidade.

Um número de classes foram já identificadas: Estudante, TrabalhadorEstudante, Docente, Administrativo e Técnico.

Todos os objetos destas classes serão caracterizadas por nome, morada e sexo. Para além disso, cada classe teria informação específica. Por exemplo, um estudante tem um número de estudante; todos os trabalhadores têm um salário.

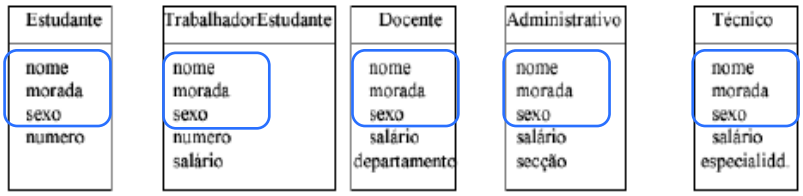
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First : *don't use a single classe!* WHY?



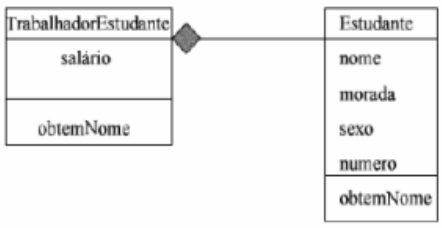
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Classes – version 0.0



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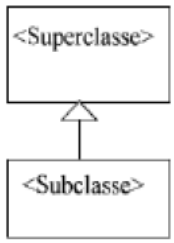
Version 0.1



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Generalization/Specialization and Inheritance

- Inheritance is a mechanism que allow to endow data and function functions of one a classe into another classe.



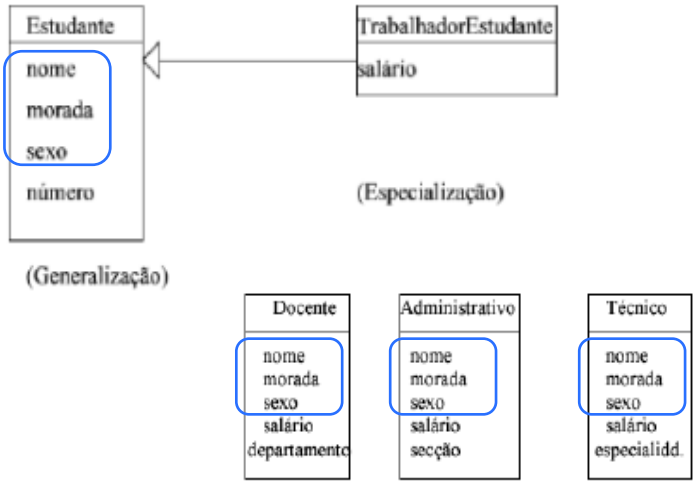
(generalizes)

(specializes)

- The subclass (or derived class) inherits all data and function members from its superclass (or base class) and can use them as its own.

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TrabalhorEstudante *é-um* Estudante

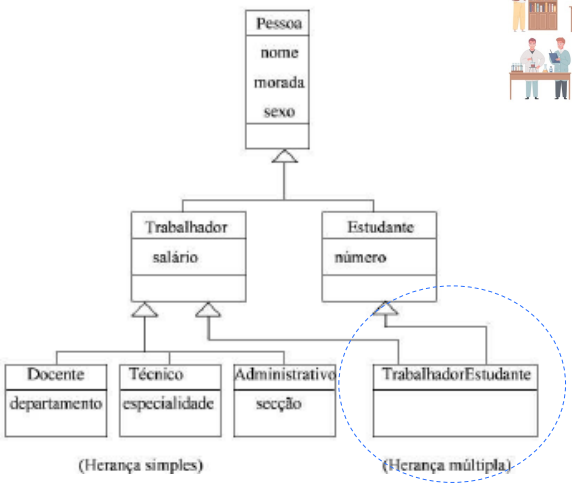


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The proposed class diagram



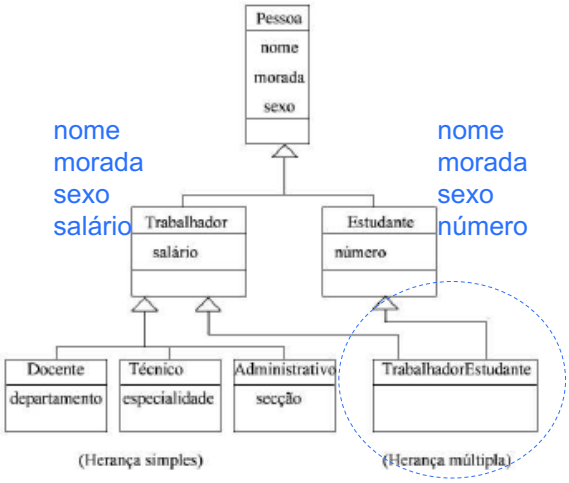
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The proposed class diagram

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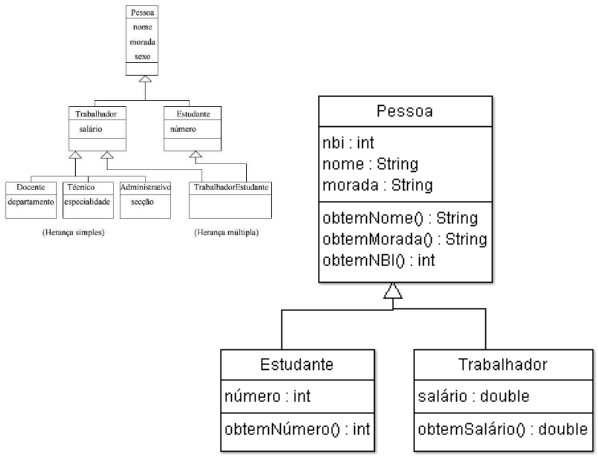
nome
morada
sexo
salário

nome
morada
sexo
número

nome
morada
sexo
salário
nome
morada
sexo
número

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Coding the diagram – first attempt



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Coding class Pessoa from the diagram

```
class Pessoa {
    private final int nbi;
    private String nome, morada;
    public Pessoa(int bi,  String n, String m) {
            nbi = bi; nome = n;  morada = m;
    };
    public int obtemNbi() {return nbi;}
    public String obtemNome() {return nome;}
    public String obtemMorada () {return morada;}
    public String toString() {
            return new String ("Bi: " + nbi + “Nome: ” + nome + "Morada:
    " + morada);
    }
}
```

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Estudante *is a* Pessoa

```
class Estudante extends Pessoa {
private final int  numero;
public Estudante(int bi, int no, String n, String m) {
        super(bi, n, m);
        numero = no;
}

public Estudante(int bi, int no, String n) {
    this(bi, no, n, "Desconhecida");
}
public int obtemNumero() {return numero; }
```

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Trabalhor *é uma* Pessoa

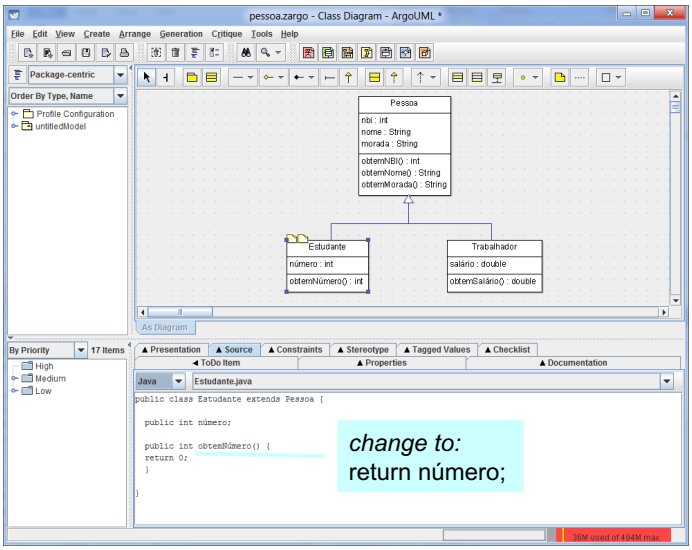
```
class Trabalhador extends Pessoa {
private float salario;

public Trabalhador(int bi, String n, String m, float f) {
    super(bi, n, m);
    salario = f;
}
public float obtemSalário(){return salario;}

public String toString() {
    String s = super.toString();
    String t = new String("Salário: " + obtemSalário());
    return s+t;
    }
}
```

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Generating base code with Argo UML



From: https://argouml-tigris-org.github.io/tigris/argouml/

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Objects initialization under inheritance

- When an object is created memory is allocated to all its instance variables, including for those inherited from all its superclasses, all of them being initialized with their default value or null.

Afterwards:

1. The superclass constructor is called
2. The initialization blocks are executed, if present
3. The body of the constructor is executed

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Example

```
Estudante e = new Estudante(222,123, "João", "Olhão");
```

```
class Estudante extends Pessoa {
  private final int  numero;

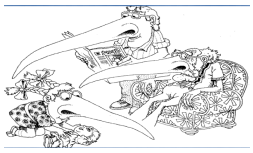
 public Estudante(int bi, int no,
   String n, String m) {
    super(bi, n, m);
    numero = no;
}
```

Memory is reserved for variables nbi, nome, morada inherited from Pessoa, and for numero. These variables have now the values: 0, null, nul, 0, respectively (why?). Afterwards:

1. The constructor Pessoa(222,"João", "Olhão") is called
2. 123 is assigned to numero

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To take away



- Generalization/Specialization – Inheritance
  - UML notation
  - Java mechanisms for inheritance
  - Initialization of objects under inheritance
  - Example 1: University people
  - Example 2: Graphic Editor

Image from: Grady Booch et al., *Object-oriented Analysis and Design with Application*, Wiley, 2007

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